

Prospective source model of the 2026 M_w 7.4 San José del Palmar (Colombia) intermediate-depth earthquake

A blind test of the MTUQ+WISP pipeline, frozen before the USGS finite fault — and the discovery of a 35-s emergent nucleation phase

Automated pipeline (17.Venezuela_2026)

Frozen 2026-08-11; USGS posted their finite fault 2026-08-14 — head-to-head in Section 10

Purpose and headline

The M_w 7.4 earthquake of 2026-08-10 beneath western Colombia (110 km deep) has, at freeze time, no published finite-fault model. We freeze a teleseismic-only source model for later head-to-head comparison. Two findings lead: (1) the event is not a Bucaramanga-nest earthquake — it belongs to the Cauca intermediate-depth cluster of the subducting Nazca slab, 407 km southwest of the nest; (2) the rupture began with a ~ 35 -s weak emergent nucleation phase before the main moment release 29 km south of the epicentre — a property that breaks naive origin-anchored inversions and had to be handled explicitly.

1 Event, tectonic setting, and the Bucaramanga question

Origin	2026-08-10T12:34:28.1 UTC, 4.844°N, 76.242°W, depth 110.3 km
Magnitude	M_w 7.4 (W-phase); $M_0 = 1.66 \times 10^{20}$ N m; centroid 120.5 km; DC 87 %
W-phase NP1	124.0/58.6/158.5
W-phase NP2	225.6/71.8/33.2 (NE–SW, steep, oblique reverse)
W-phase centroid	4.585°N, 76.161°W (29 km S of epicentre), 12:35:02.6 = origin + 34.5 s

Bucaramanga nest? No. The nest — the world’s densest intermediate-depth seismic nest — sits at $\sim 6.82^\circ\text{N}$, 73.15°W , 140–160 km depth beneath Santander, NE Colombia. This event is 407 km to the southwest at 4.84°N , 76.24°W , 110–120 km deep: the Cauca intermediate-depth cluster of the subducting Nazca slab (cf. Chang et al. 2019), a different slab segment with its own seismicity. It ranks among the largest instrumental events of that segment.

2 Data

120 three-component stations (316 channels), 30–90°, maximum azimuthal gap 47° (the Pacific). MTUQ used an azimuth-balanced 36-station response-corrected subset with the deep-event bands (body 25–60 s, mantle surface waves 200–450 s). Both nodal planes extend below the 126.5-km

WISP surface-wave GF cap, so the finite-fault plane comparison is body-only on both planes — fair by construction (the San Pedro de Atacama rule).

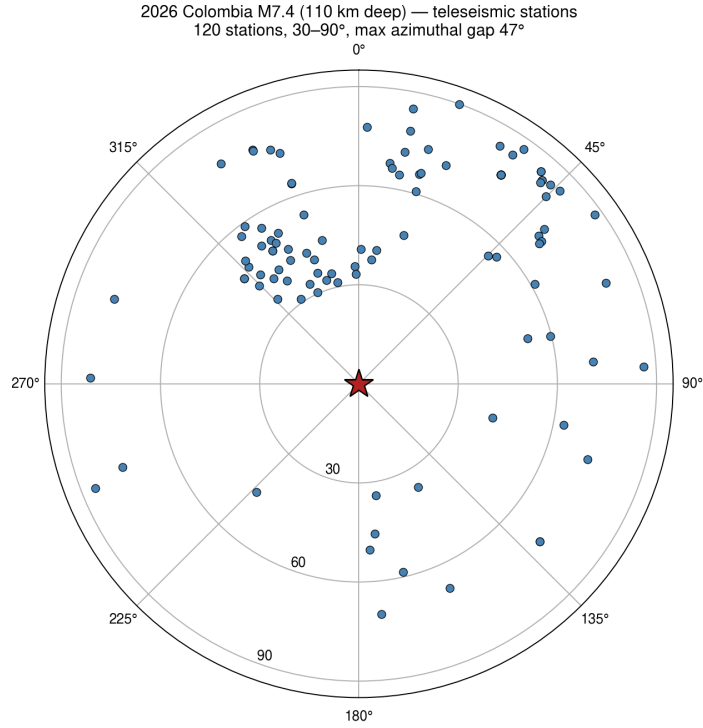


Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 The emergent nucleation phase — and why origin-anchored inversions fail

The observed P wavetrains begin weakly and emergently; the main impulsive packet arrives only $\sim 45\text{--}55$ s after origin (its pP/sP depth-phase echoes at $\sim 75\text{--}90$ s). Independently, the W-phase centroid time lags the origin by 34.5 s and sits 29 km south. The implied nucleation-to-asperity apparent velocity is $\sim 0.8 \text{ km s}^{-1}$ — far below any physical rupture speed — so a finite fault nucleating at the cataloged hypocentre at the cataloged time cannot place the moment where the data demand it. Our first (origin-anchored) inversion demonstrated the failure exactly: slip piled at the fault edge (86% of peak in an edge row — a violated containment rule), the STF peaked at an unreachable 40 s, and the synthetics fit only the long-period envelope. Origin-cut MTUQ windows likewise missed the main energy (M_w pinned at the grid edge).

The fix (the operational Yeck-style relocation): re-anchor the entire analysis — window cutting, Green’s functions, fault geometry, nucleation — at the W-phase centroid in space and time. The 35-s precursor then lies outside the modelling window (its moment is a few per cent at most), and the main rupture is imaged about its true origin.

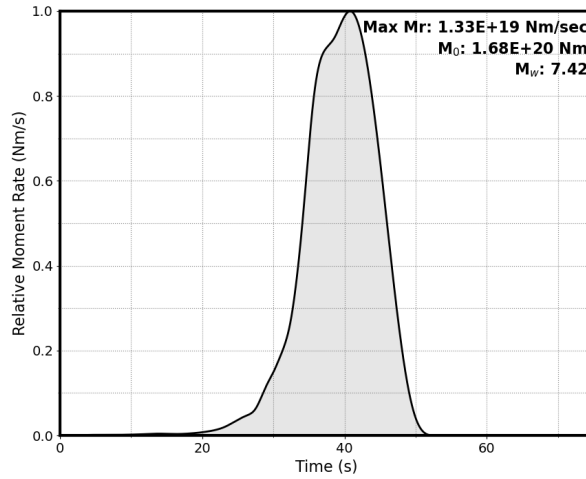


Figure 2: The INVALID origin-anchored model’s STF, shown as the diagnostic: the peak at ~ 40 s is kinematically unreachable from the hypocentre, so the inversion smeared slip to the fault edges.

4 Finite fault (WISP, centroid-anchored): both planes, **shift_match**, body-only

plane (W-phase labels)	misfit (post-shift)	edge slip top/bottom	peak slip
226/72/33 (preferred)	0.1363	1 % / 3 %	1.63 m
124/59/159 conjugate	0.1485	40 % / 2 %	1.12 m

Table 1: An 8.2 % margin plus the conjugate’s top-edge violation = a near-decisive plane call for the NE–SW steep oblique-reverse plane — exactly where the rupture-size criterion (Calama 1.6 % tie at 10 km; SPdA 10 % decisive at 50 km) predicts discrimination turns on. WISP’s directory labels are swapped vs USGS (WISP NP1 = W-phase NP2), as at Kumamoto. Recovered $M_0 = 1.68 \times 10^{20}$ N m (W-phase 1.66).

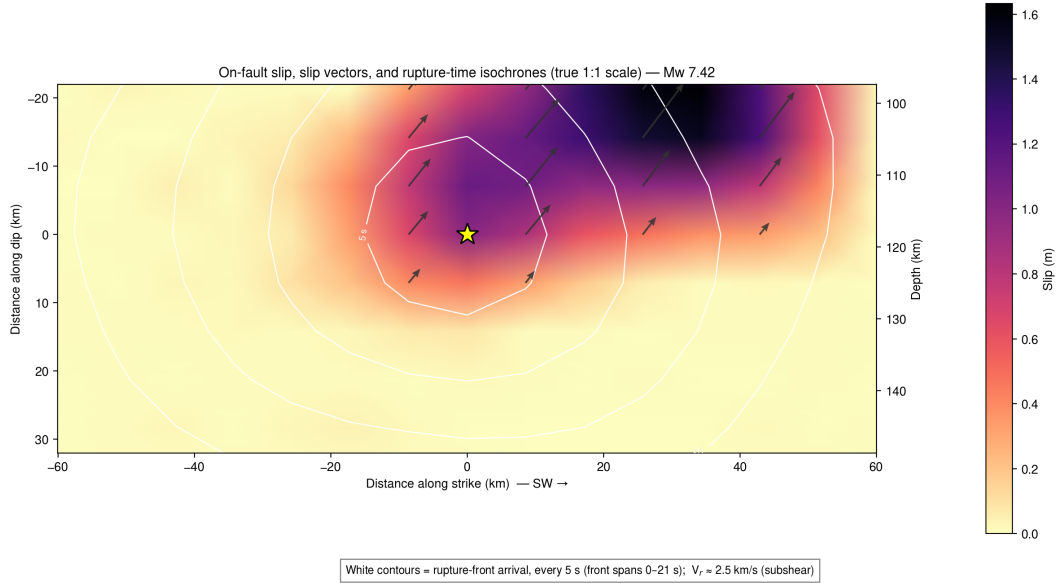


Figure 3: Preferred plane (226/72/33), true scale: a single dominant asperity ~ 17 km SSW of the centroid nucleation (~ 45 km SSW of the epicentre), up-dip, slip at 85–125 km depth, peak 1.63 m, cleanly contained (edge slip 1–3%). Effective area $2695 \text{ km}^2 \rightarrow$ radius 29 km $\rightarrow$ stress drop ≈ 2.9 MPa.

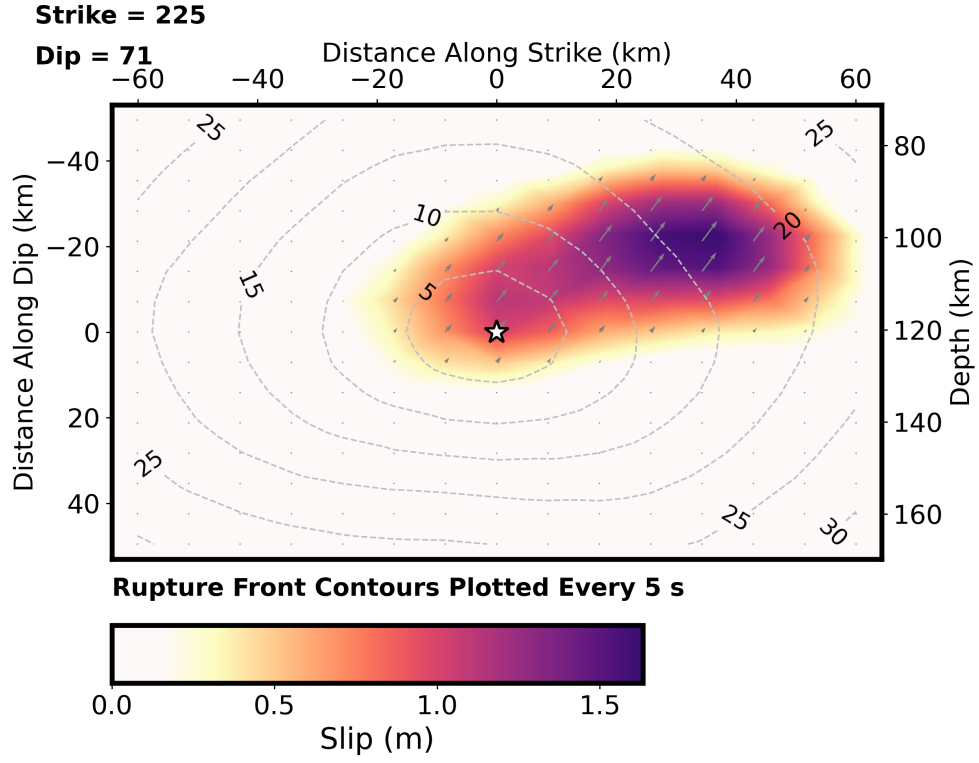


Figure 4: Same model with 5-s rupture-front isochrones: main-phase duration $T_{5-95} \approx 18.7$ s (times relative to the centroid nucleation; the ~ 35 -s precursor precedes this and is below our band).

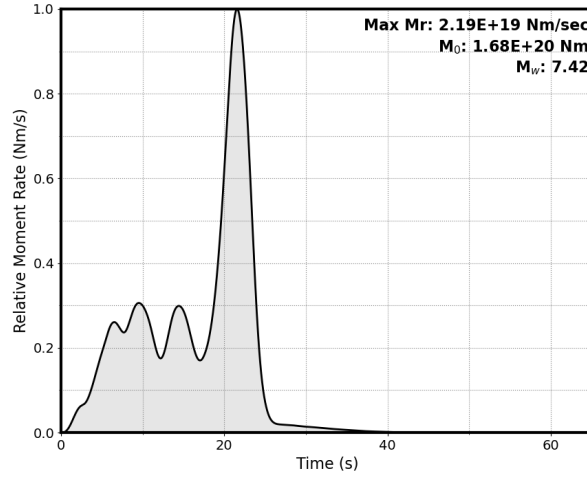


Figure 5: Main-phase moment-rate function (centroid-anchored, preferred plane).

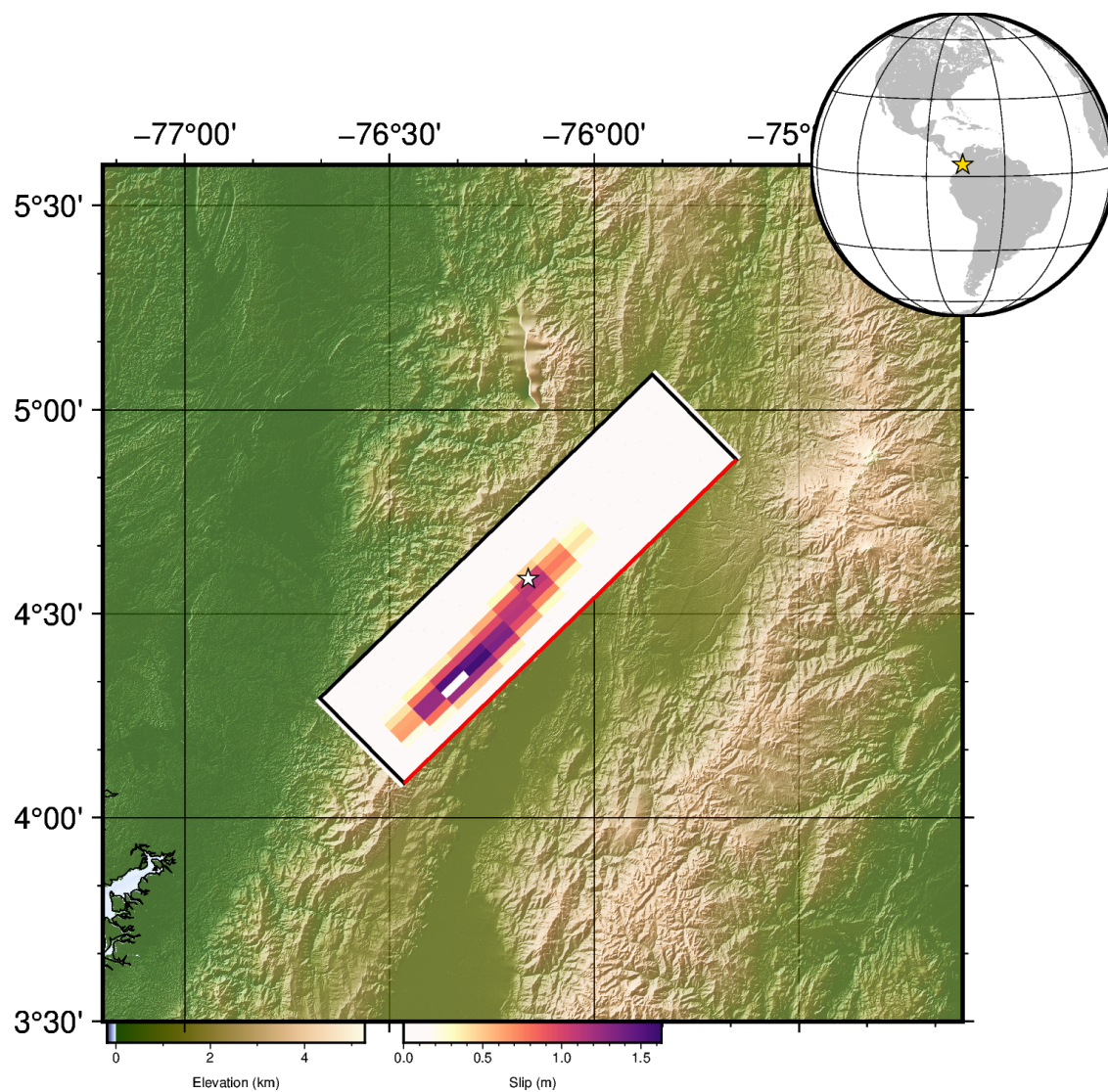


Figure 6: Map view of the preferred-plane slip.

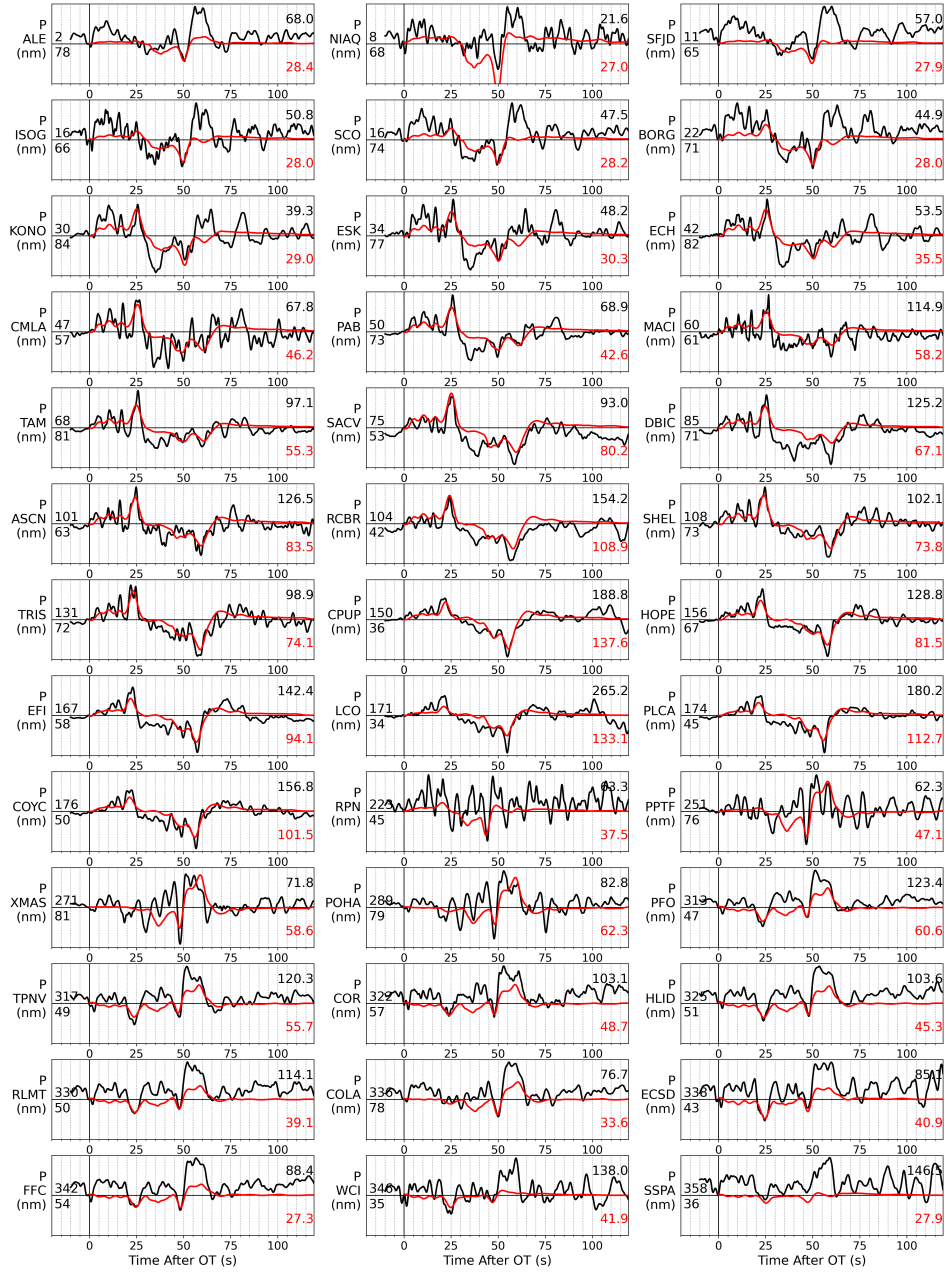


Figure 7: Teleseismic P fits (preferred plane, shifted): the main packet and its depth phases are now aligned and fit.

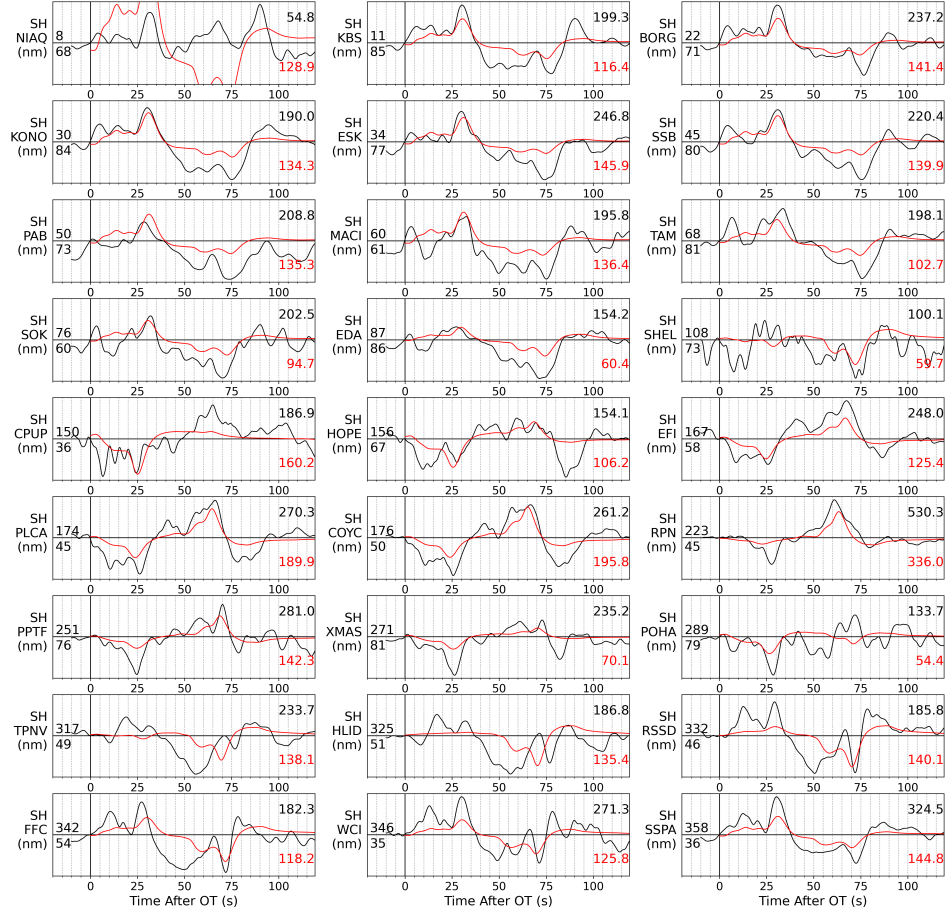


Figure 8: Teleseismic SH fits (preferred plane, shifted).

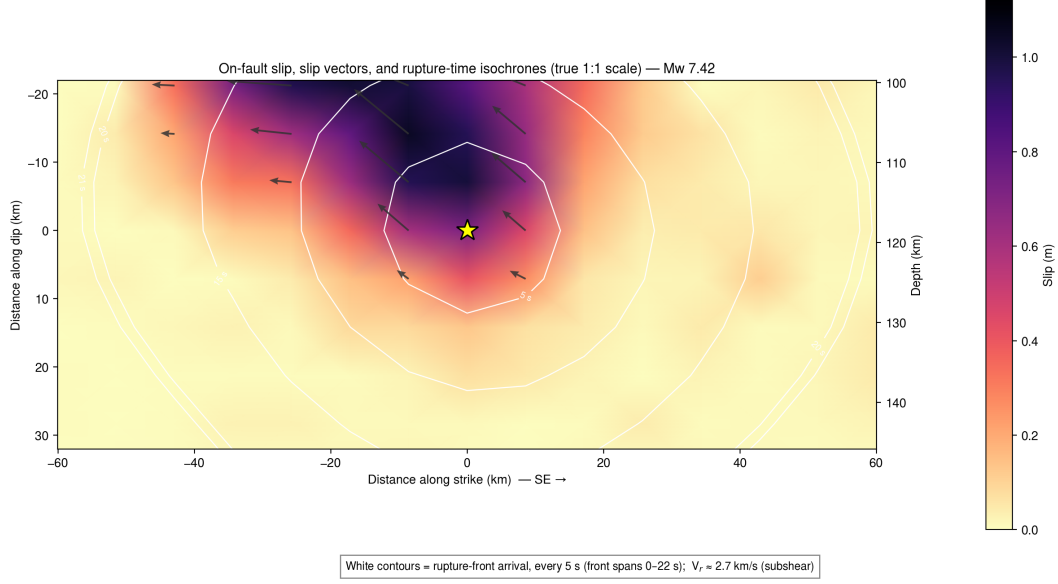


Figure 9: Conjugate plane (124/59/159) for comparison: 8.2 % worse fit and 40 % of peak slip at its top edge.

5 Moment tensor (MTUQ): the point source breaks on this complex source

variant	mechanism	M_w	Kagan vs W-phase
joint DC @120.5 km	202/81/−65	7.10 (grid edge)	95.3°
body P+SH @120.5 km	220/61/−52	7.10 (grid edge)	83.6°
body depth scan (60–150 km)	minimum at 70–90 km	7.05–7.25	unstable 28.6–93°

Table 2: The body-wave depth curve bottoms at 70–90 km — shallower than the W-phase centroid — and the mechanism is stable nowhere except ~ 90 km, where it snaps to 225/71/4 (28.6°). A 40-km-offset late asperity, a 35-s precursor and a 40-km-tall slip zone violate the point-source approximation. The robust element — strike ~ 207 – 225° — matches the preferred finite-fault plane. Pipeline rule: for complex sources, take mechanism and depth from the W-phase and spend the teleseismic data on the finite fault.

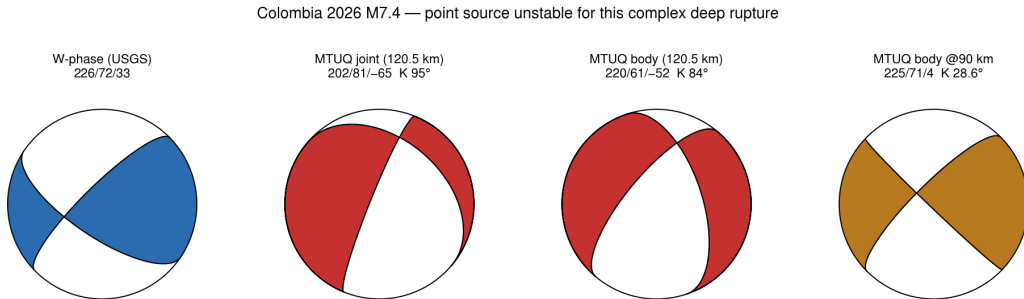


Figure 10: W-phase reference vs MTUQ attempts.

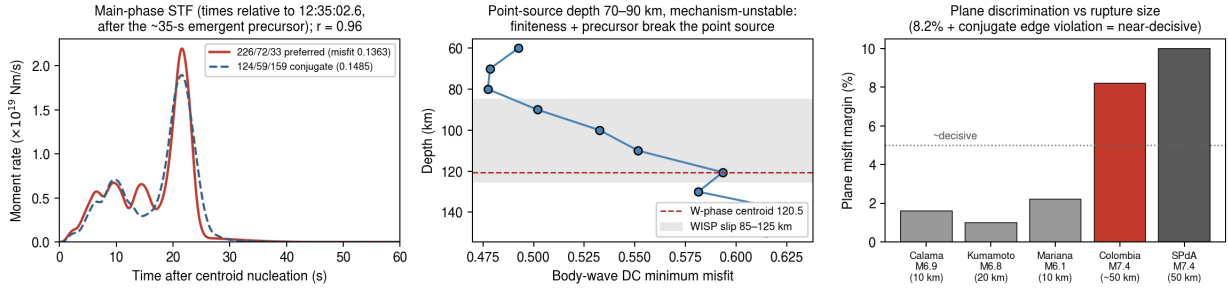


Figure 11: Left: the main-phase STF is stable across the two assumed planes. Centre: the point-source depth curve vs the finite-source depth extent — the point source is biased shallow by finiteness. Right: the plane-discrimination margin in the pipeline’s rupture-size context.

6 Directivity: two-stage, unilateral southward + up-dip

(A correction made within the same pre-USGS freeze window: the first push described the asperity as “NE along strike” — an axis-misread of the WISP plot; the geographic computation is authoritative.)

1. Model-independent stage 1: epicentre $\rightarrow$ W-phase centroid = 30 km toward azimuth 163° (SSE), 110 $\rightarrow$ 120.5 km, in ~ 34.5 s — apparent 0.9 km s^{-1} (the emergent nucleation phase).
2. Preferred plane stage 2: the early $\rightarrow$ late moment centroid moves 27 km toward azimuth 220° (SW) while shallowing 110 $\rightarrow$ 101 km; 75 % of the moment lies forward of the nucleation vs 24 % behind — predominantly unilateral, along-strike SW plus up-dip.
3. Robustness: the up-dip component appears on both planes; the conjugate’s horizontal direction (northward) contradicts the model-independent stage-1 offset, further disfavoring that already-rejected plane.

7 Tectonic context: background seismicity, the Caldas tear, and tomography

Background: 1,981 ComCat events ($M \geq 4$, depth ≥ 60 km, 1980–2026) and the CARMA finite-frequency dV_p model of Cornthwaite et al. (2021; IRIS EMC CARMA_nwSA_dVp_Tomo), the same array behind Sun et al. (2022). (The EMC file labels its units “percent” but stores fractions; values are rescaled here — $\max |dV_p| \approx 9.9\%$, matching the publications.)

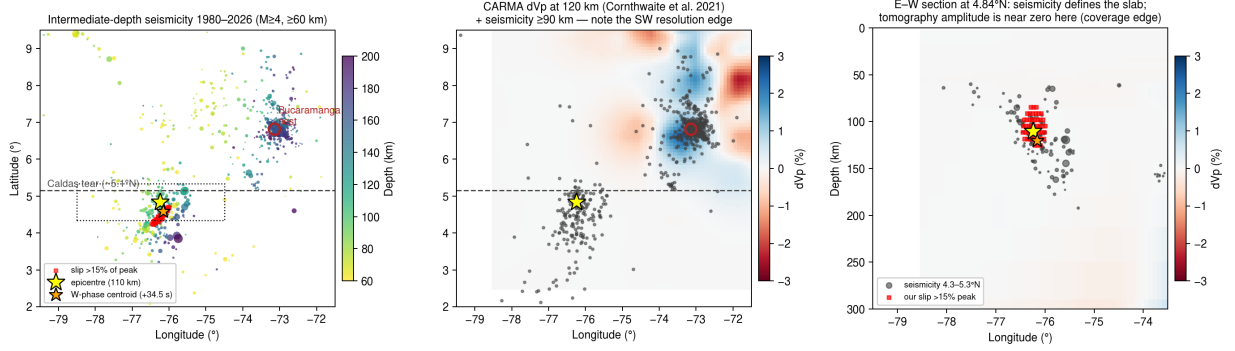


Figure 12: Left: intermediate-depth seismicity; the rupture (red) lies inside the NNE-trending Cauca cluster, which terminates at the Caldas tear ($\sim 5.15^\circ\text{N}$); the Bucaramanga nest is the dense knot 407 km NE. Centre: CARMA dV_p at 120 km — the strong fast anomaly underlies Bucaramanga (the Caribbean slab of Sun et al. 2022); amplitude at our event is $\sim 0.01\%$ (SW resolution edge). Right: E–W section at 4.84°N — the slip patch (85–125 km) sits exactly on the east-dipping seismicity band that defines the local Nazca slab.

Four statements follow. (1) The rupture is the Cauca cluster breaking: the slip zone coincides with the cluster’s seismicity band in map view and in depth, with the hypocentre at its northern/updip edge and rupture propagating SSW into the cluster interior, away from the tear. (2) The Caldas tear bounds the system: the Cauca band terminates at $\sim 5.15^\circ\text{N}$ and a marked gap in deep seismicity ($5.2\text{--}6.5^\circ\text{N}$) separates it from the Bucaramanga domain — consistent with the event’s along-strike-compression (map-view strike-slip) mechanism adjacent to the tear. (3) Bucaramanga is a different system: per Sun et al. (2022) the nest lies in the Caribbean slab (the fast anomaly in the centre panel), so nest and event are plausibly different subducted plates, not merely different segments. (4) Tomographic coverage, honestly: the CARMA model’s domain nominally contains the event but its amplitude there is at the resolution edge; the local slab is defined by seismicity, not tomography.

On-fault view: projected background seismicity and prior M6+ mechanisms

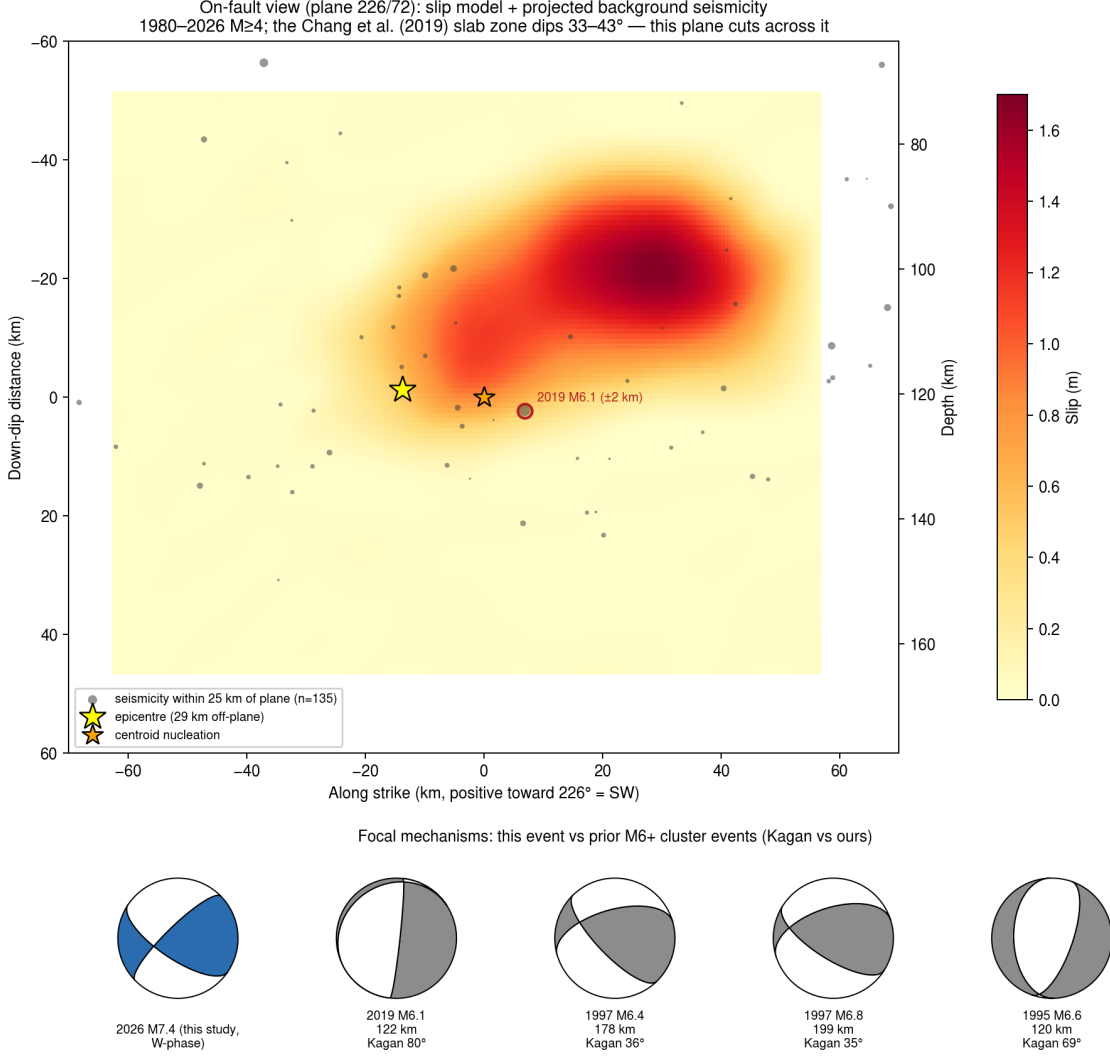


Figure 13: Background seismicity ($M \geq 4$, 1980–2026) within 25 km of the preferred plane, projected into along-strike/down-dip coordinates over the slip model; below, prior M6+ cluster mechanisms with Kagan angles vs our W-phase mechanism.

Four observations. (1) The main rupture nucleated where the cluster’s most recent M6 broke: the 2019 M6.1 Versailles hypocentre projects ~ 2 km off-plane, ~ 7 km from our centroid nucleation — yet its mechanism (237/8/−40) is 80° from ours: the same volume fails on differently oriented structures, the Chang et al. (2019) mechanism diversity expressed at M6+. (2) Only the deep 1997 Roncesvalles pair (M6.8/199 km, M6.4/178 km) resembles our mechanism (Kagan 35–36°); the in-cluster 1995 M6.6 Neira was normal-type (69°). (3) The rupture plane cuts across the slab: the relocated slab zone dips 33–43° (Chang et al. 2019) vs our 72° plane — a steep inherited structure within the slab, as at San Pedro de Atacama. (4) No prior $M \geq 4$ concentration outlines the main asperity: the 1.6-m patch was comparatively quiet beforehand (ComCat completeness caveat).

Rupture propagation snapshots

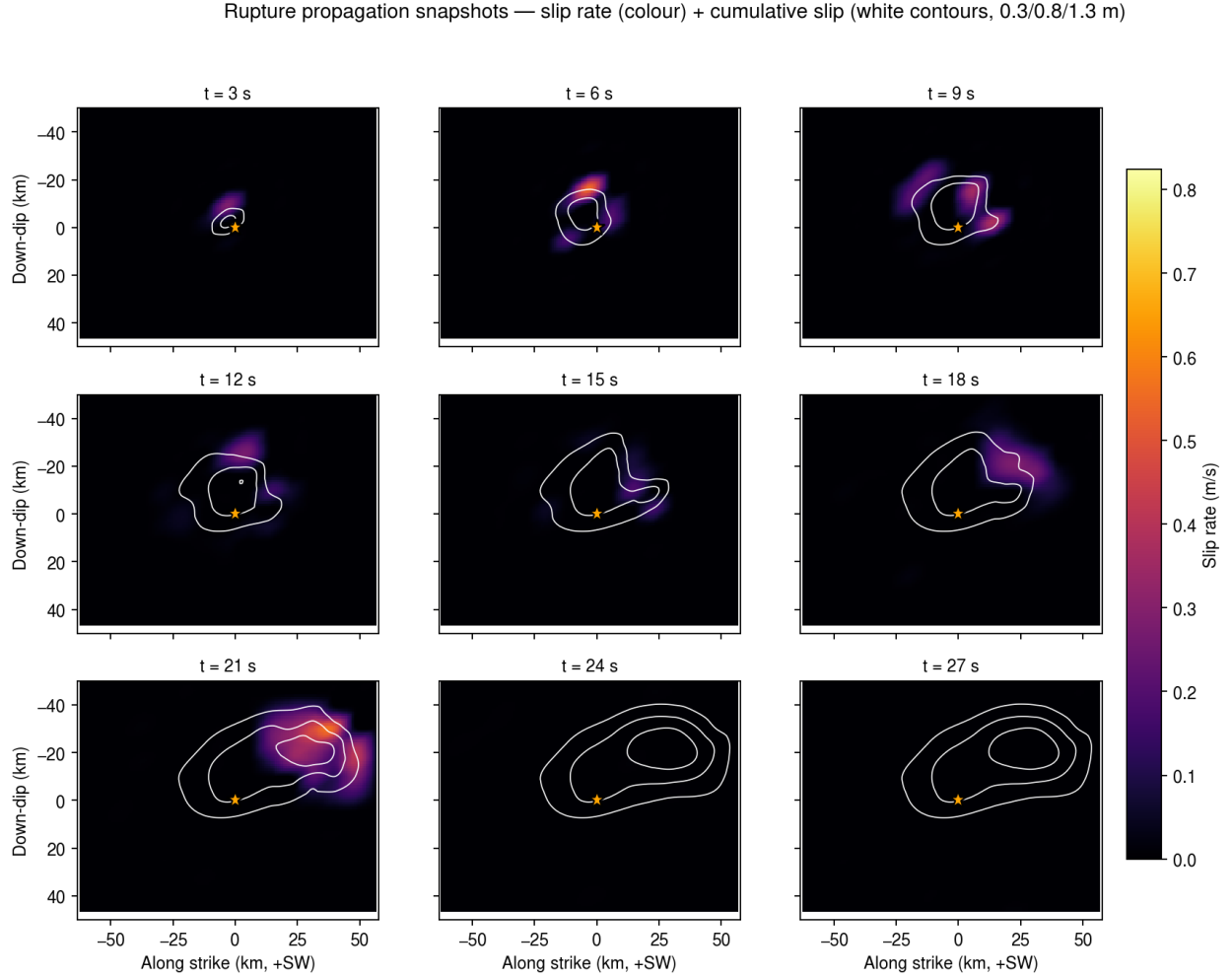


Figure 14: Slip-rate snapshots reconstructed from the inverted rupture times and asymmetric-cosine slip-velocity functions; white contours: cumulative slip (0.3/0.8/1.3 m). Growth around the nucleation (3–9 s), the front sweeping SW + up-dip (12–18 s), the main asperity failing at ~18–21 s (the STF peak), arrest by ~24 s — the unilateral directivity frame by frame. An animated time-lapse is provided as `colombia_m74_rupture.gif` alongside this report.

8 Frozen prospective prediction

- Fault plane: 226/72/33 (NE–SW, steep, oblique reverse) — 8.2 % margin plus the conjugate’s edge violation.
- Two-stage rupture: a ~35-s weak emergent nucleation from the cataloged hypocentre, then the main asperity ~45 km SSW of the epicentre, up-dip, 85–125 km depth; peak slip ≈ 1.6 m; main-phase $T_{5-95} \approx 19$ s; $\Delta\sigma \approx 2.9$ MPa.
- M_0 1.68×10^{20} N m (M_w 7.42), matching the W-phase to 1 %.

- Scorecard when USGS posts: plane choice; whether they relocate the hypocentre/nucleation; asperity position (SSW + up-dip); depth extent; peak slip; duration.

Caveats. Teleseismic-only; the precursor’s own moment and location are below our band and not imaged; the mechanism statement rests on the W-phase (MTUQ instability documented above); body-only fits on both planes (surface-wave GF cap).

9 Head-to-head with the USGS finite fault (posted 4 days after the freeze)

quantity	ours (frozen)	USGS
fault plane	226/72/33	225.57/71.78/33.23 — identical
M_0 (M_w)	1.68×10^{20} (7.42)	1.72×10^{20} (7.424)
main asperity	~ 44 km S of epicentre, ~ 105 km	31 km SSW (az 209°), 100 km
slip depth ($>15\%$ peak)	85–125 km	91–108 km
directivity	unilateral SW, 27 km @ 220°	unilateral SW, 46 km @ 225°
main-phase duration	~ 19 s	~ 24 s
main-phase V_r (same metric)	2.52 km s^{-1}	2.34 km s^{-1}
nucleation handling	re-anchored at W-phase centroid	also relocated (24 km NE, 125 km) + silent crawl
stage-1 moment	excluded, bounded $\leq$ few %	0.2 % in the first 25 s
peak slip	1.63 m	3.98 m
$\Delta\sigma$ (effective-area)	2.9 MPa	8.1 MPa

Table 3: Every physical headline of the frozen prediction verified, including the methodological one: USGS could not nucleate at the cataloged hypocentre either. Their model represents the two-stage source as a silent slow crawl (header avV_r 1.28 km s^{-1}) from a relocated initiation point; ours excludes stage 1 by re-anchoring. Their near-zero early moment independently confirms our stage-1 bound. Honest differences: peak slip ($\times 2.4$) and stress drop track their smaller subfaults/weaker smoothing; the initiation point — catalog vs their NE relocation vs the W-phase centroid — remains the genuinely ill-determined element, while the moment-release location is now triple-consistent.

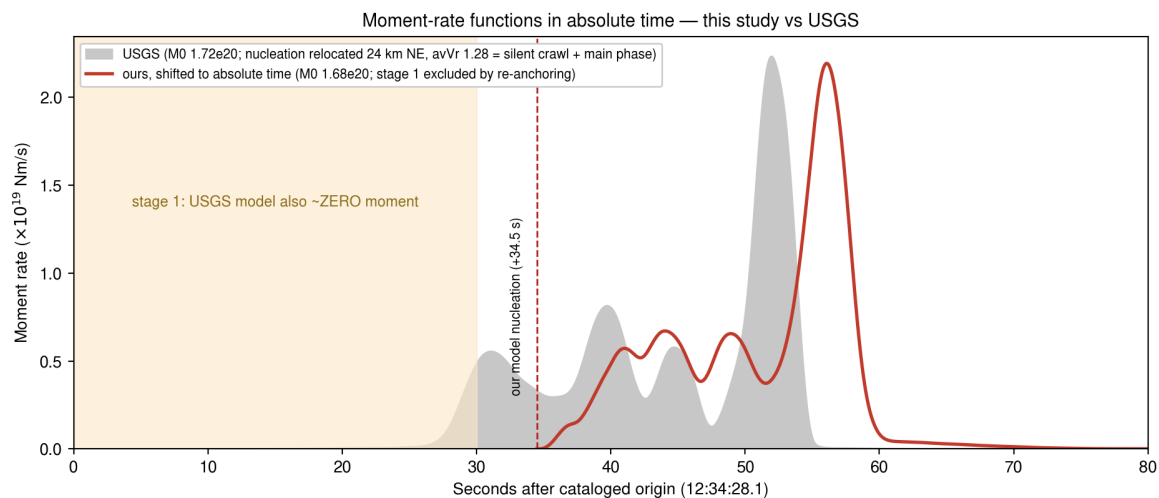


Figure 15: Moment rate in absolute time: USGS (grey) and ours shifted to absolute time (red). Both models carry essentially no moment during stage 1.

Map-view slip distribution — this study vs USGS

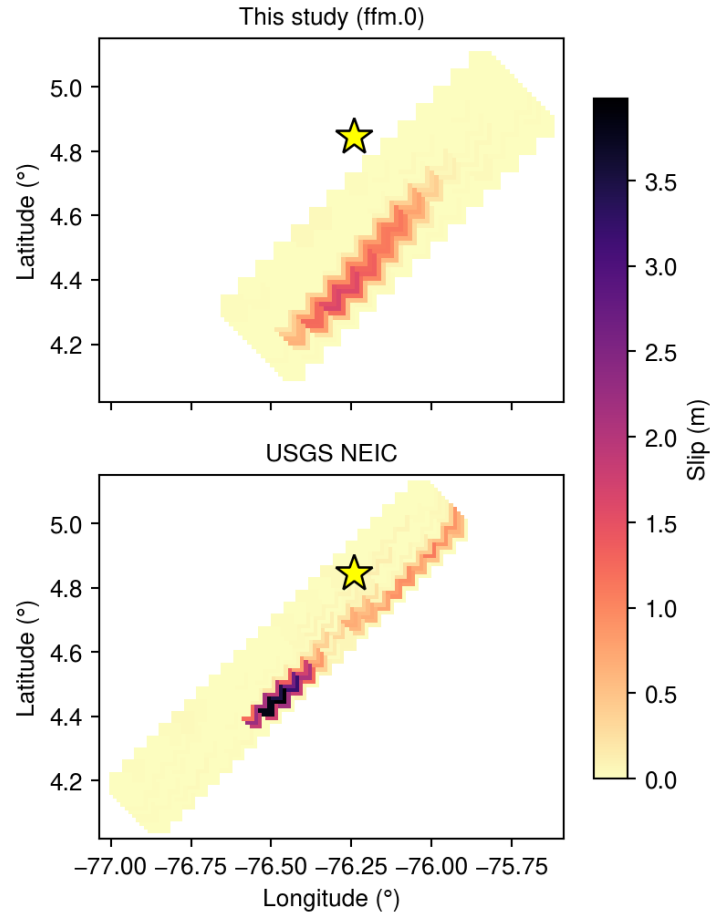


Figure 16: Map-view slip, this study vs USGS.

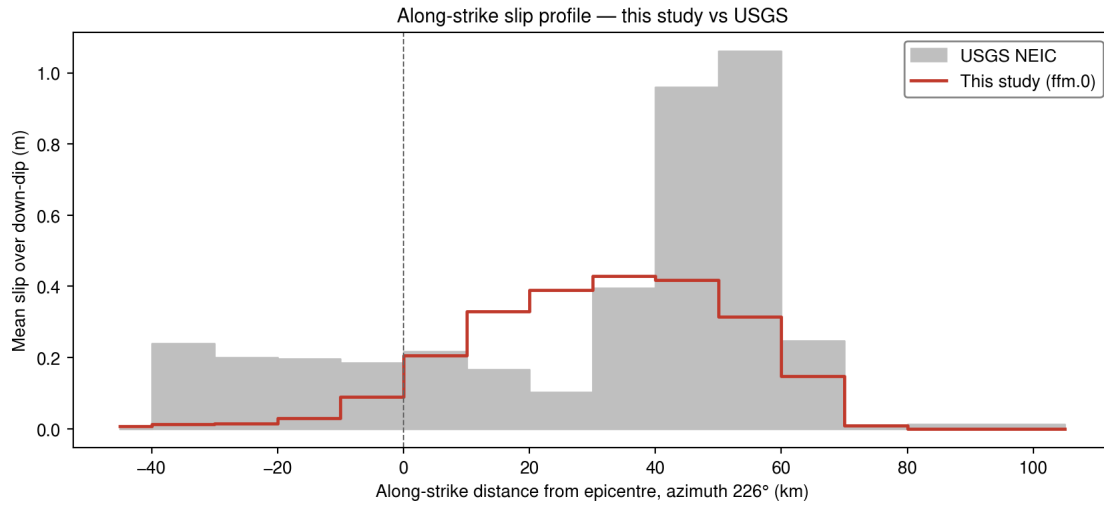


Figure 17: Along-strike slip profiles (azimuth 225.6° through the epicentre).